Task 6: Session 6 (AIML) - Assignment Questions

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1. **Creating Lists**

   Write a program to create the following lists using different methods:

   - A list of 5 integers directly.
   - An empty list using both [] and list().
   - A list containing mixed data types (int, string, float, boolean).
   - A list of 7 zeros using repetition (*).

   Print all lists with clear labels and add comments explaining each creation method.

   Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
List Of 5 Integers: [10, 20, 30, 40, 50]
Empty List Using []: []
Empty List Using list(): []
List With Mixed Data Types: [42, 'hello', 3.14, True]
List Of 7 Zeros: [0, 0, 0, 0, 0, 0, 0]
PS D:\Mountreach_ITR_2026>
```

2. **Indexing & Negative Indexing**

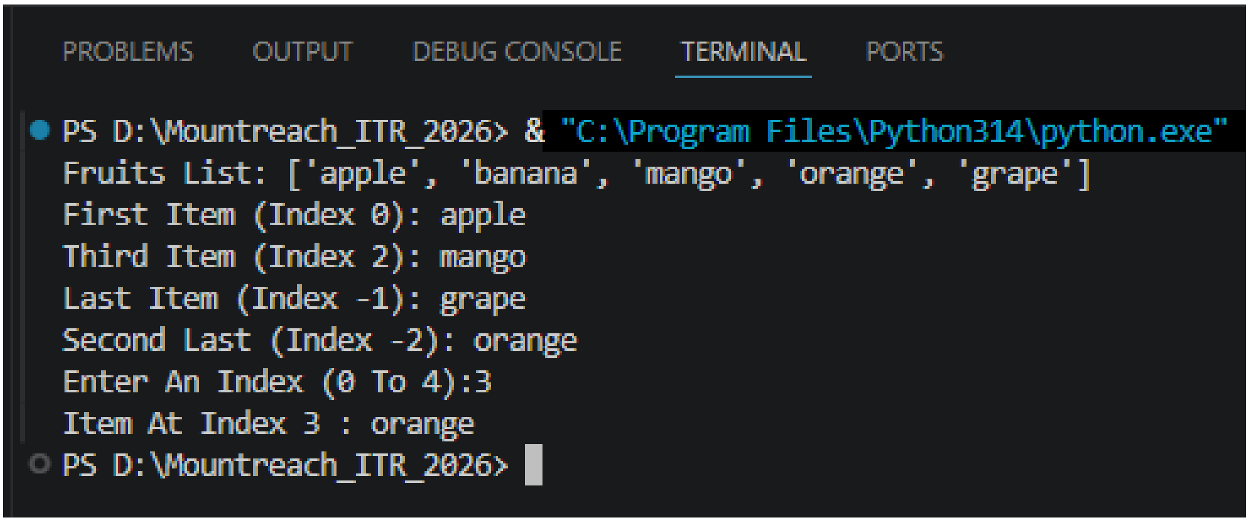
Create a list: fruits = ["apple", "banana", "mango", "orange", "grape"]

Write code to print:

- First item
- Third item
- Last item (using negative index)
- Second last item (using negative index)

Take a number as input from the user and print the item at that index (handle basic cases only).

Output:



3. Slicing Lists

Given the list: numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

Using slicing, print:

- First 4 elements
- Last 3 elements
- Elements from index 2 to 7
- Every second element
- The entire list in reverse order

Add comments explaining slicing syntax.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers List: [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
First 4 Elements: [10, 20, 30, 40]
Last 3 Elements: [80, 90, 100]
Index 2 To 6: [30, 40, 50, 60, 70]
Every Second Element: [10, 30, 50, 70, 90]
Reverse The List: [100, 90, 80, 70, 60, 50, 40, 30, 20, 10]
PS D:\Mountreach_ITR_2026>
```

4. Modifying Lists

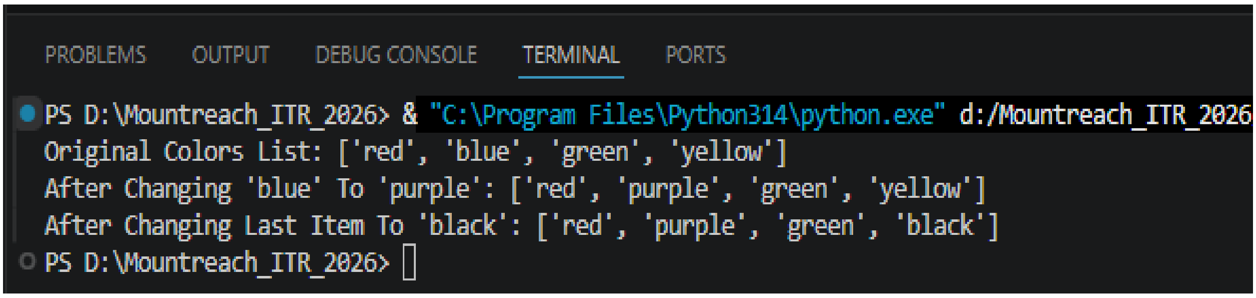
Create a list: colors = ["red", "blue", "green", "yellow"]

Perform the following:

- Change "blue" to "purple" using indexing.
- Change the last item to "black".

Print the list after each change.

Output:



5. append() & insert()

Write a program that starts with an empty list.

- Use append() to add 5 different city names one by one.
- Use insert() to add one more city at position 2.
- Print the final list.

Take user input for at least 2 cities.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A City Name:Pune
Enter Another City Name:Indore
After 5 Appends:
 ['Mumbai', 'Delhi', 'Bangalore', 'Pune', 'Indore']
After Inserting 'Hyderabad' At Position 2:
 ['Mumbai', 'Delhi', 'Hyderabad', 'Bangalore', 'Pune', 'Indore']
PS D:\Mountreach_ITR_2026>
```

6. remove(), pop(), & clear()

Create a list: items = [10, 20, 30, 20, 40, 50]

Perform and print after each operation:

- Remove the first occurrence of 20 using remove().
- Remove the item at index 3 using pop() and print the removed value.
- Remove the last item using pop().
- Clear the entire list using clear().

Add comments explaining the difference between remove() and pop().

Output:

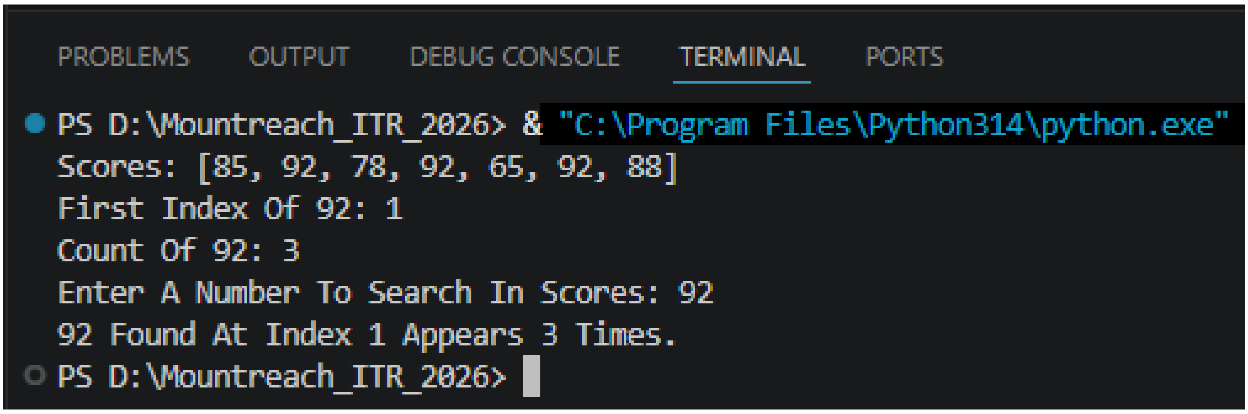
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original: [10, 20, 30, 20, 40, 50]
After Remove(20): [10, 30, 20, 40, 50]
After pop(3) Removed Value: 40 List Now: [10, 30, 20, 50]
After pop(): [10, 30, 20]
After clear(): []
PS D:\Mountreach_ITR_2026>
```

7. index() & count()

Create a list: scores = [85, 92, 78, 92, 65, 92, 88]

- Find and print the index of the first occurrence of 92.
- Count and print how many times 92 appears.
- Take a number as input from the user and check if it exists in the list. If yes, print its index and count.

Output:



8. sort() & reverse()

Create a list: marks = [45, 78, 65, 90, 34, 82, 71]

- Sort the list in ascending order and print it.
- Sort the list in descending order and print it.
- Reverse the original list (without sorting) and print it.

Add comments on the difference between sort() and reverse().

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Original Marks: [45, 78, 65, 90, 34, 82, 71]
Ascending Order: [34, 45, 65, 71, 78, 82, 90]
Descending Order: [90, 82, 78, 71, 65, 45, 34]
Reversed (No Sort): [71, 82, 34, 90, 65, 78, 45]
PS D:\Mountreach_ITR_2026>
```

9. extend() vs append()

Create two lists:

list1 = [1, 2, 3]

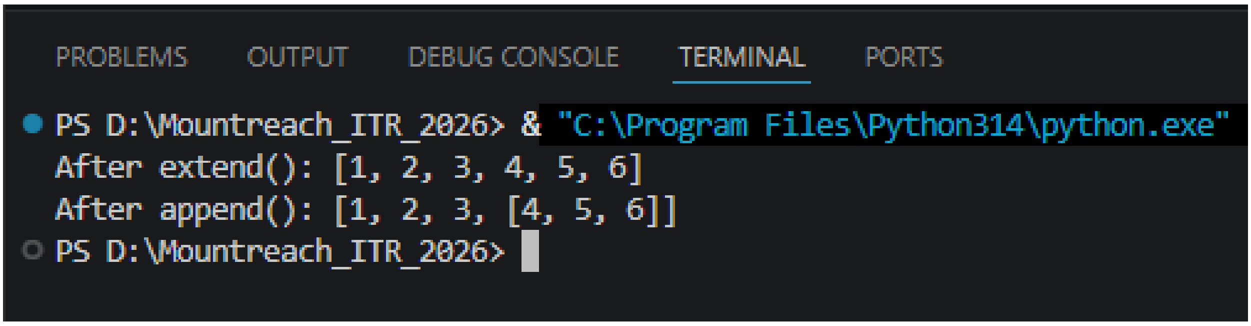
list2 = [4, 5, 6]

Demonstrate:

- Using extend() to add list2 to list1.
- Using append() to add list2 to a copy of list1.

Print both results and explain the difference in comments.

Output:



10.Mini Project - Combined Concepts

Create a **Student Marks Manager** program using lists:

- Take 5 subject marks as input from the user and store them in a list.
- Print the list.
- Add one more subject mark using append().
- Print the highest and lowest marks (using built-in functions).
- Sort the marks in descending order.
- Calculate and print the average marks.
- Use len() to show total number of subjects.

Use proper comments, meaningful variable names, and control structures where needed.

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter Marks For Subjects:98
Enter Marks For Subjects:67

Initial Marks List: [87, 45, 23, 98, 67]

Enter Marks For The Extra Subject:84

After Adding Extra Subject: [87, 45, 23, 98, 67, 84]

Highest Marks: 98
Lowest Marks: 23

Marks In Descending Order: [98, 87, 84, 67, 45, 23]

Average Marks: 67.33333333333333

Total Number Of Subjects: 6
PS D:\Mountreach_ITR_2026>
```